

Retro-Causality within the Triadic Cosmos

Future-Consistency Normalization in Dual-Time Closure-Boundary Systems

Joachim De Witte

August 2026

©2026 Joachim De Witte. All rights reserved.

Published under the Creative Commons BY-NC-ND 4.0 license.

Abstract

Retro-causal phenomena have long challenged physical theory, appearing in quantum experiments, holographic dualities, and cosmological attractor behaviour. Standard interpretations typically invoke non-locality, determinism, or block-universe ontology, yet none of these frameworks provide a coherent account that preserves causal direction while explaining why future boundary conditions appear to influence present closure states. This paper develops a structural resolution within the *Triadic Cosmos* ecosystem by extending the Universal Closure-Boundary Normalization Principle (UCBNP) to dual-time systems.

We show that retro-causal signatures arise naturally when a finite local closure evolves in material time t_{mat} while interacting with a holographic boundary canon that evolves in electromagnetic time t_{EM} . Because the boundary canon is time-invariant, closure evolution constitutes a recursive normalization toward a future-consistent attractor. This produces the appearance of retro-causality without requiring time reversal, determinism, or global initial conditions. Instead, future-consistency normalization emerges as a structural property of dual-time closure-boundary systems.

We demonstrate that this architecture provides a unified explanation for several existing puzzles: causal-structure anomalies in AdS/CFT bulk reconstruction, retro-causal signatures in delayed-choice quantum-eraser experiments, predictive processing in biological cognition, and attractor formation in cosmological bulk-boundary dynamics. In each case, the apparent influence of future events is reinterpreted as closure normalization toward a time-independent boundary canon.

Retro-causality within the Triadic Cosmos is therefore not a violation of causal order, but an emergent property of systems whose evolution is governed by dual temporal scales and holographic projection. This establishes future-consistency normalization as a domain-independent mechanism underlying retro-causal behaviour across physics, cognition, and artificial systems.

1 Introduction

Retro-causal phenomena occupy a peculiar position in contemporary science. They appear repeatedly across quantum experiments, holographic dualities, cosmological dynamics, and biological cognition, yet no existing theoretical framework explains them without invoking either non-locality, determinism, or block-universe ontology. Delayed-choice quantum-eraser experiments seem to allow future measurement settings to influence past interference patterns; bulk reconstruction in AdS/CFT suggests that boundary data can determine bulk events outside their naive causal domain; predictive-processing models of cognition imply that perceptual content is shaped by future-oriented priors; and cosmological attractors exhibit behaviour that appears guided by boundary conditions not yet realized. These signatures are typically treated as paradoxes or mathematical curiosities rather than as manifestations of a deeper structural principle.

The *Triadic Cosmos* ecosystem provides such a principle. In earlier work, the Universal Closure-Boundary Normalization Principle (UCBNP) established that systems composed of a finite local closure and a

high-dimensional holographic boundary evolve through a recursive triadic cycle of information, energy, and geometry. Crucially, closures evolve in material time t_{mat} , a sequential and history-dependent progression, while boundaries evolve in electromagnetic time t_{EM} , a time-invariant projection domain whose internal encoding does not reorganize during interaction. This dual-time architecture implies that closures normalize toward a boundary canon that is effectively timeless. As a result, closure states may exhibit anticipatory structure that appears retro-causal when interpreted within a single-time ontology.

This paper develops the consequences of dual-time closure-boundary dynamics for the interpretation of retro-causal phenomena. We introduce *future-consistency normalization*: the process by which a closure reorganizes its internal geometry to align with a boundary canon that does not evolve in material time. Because the boundary canon is time-independent, the closure’s recursive evolution can produce states that appear influenced by future boundary projections, even though no physical time reversal occurs. Retro-causality is therefore not a fundamental feature of nature, but an emergent property of systems whose evolution is governed by dual temporal scales and holographic projection.

We show that this mechanism provides a unified structural explanation for several existing puzzles. In quantum physics, delayed-choice and quantum-eraser experiments become coherent once measurement outcomes are interpreted as closure-internal normalization toward a time-invariant boundary canon. In holographic duality, causal-structure anomalies in AdS/CFT arise naturally from bulk closures normalizing toward boundary encodings that do not evolve in material time. In cosmology, attractor behaviour reflects bulk matter reorganizing toward boundary fields that function as canonical constraints. In cognition, predictive processing emerges as biological closure normalization toward environmental and cultural boundary structure.

By situating these phenomena within a single dual-time closure-boundary architecture, this paper argues that retro-causal signatures across physics, cognition, and artificial systems are not paradoxes but structural consequences of future-consistency normalization. Retro-causality within the Triadic Cosmos is thus reinterpreted as a domain-independent emergent behaviour arising from the interaction of finite closures with timeless holographic boundaries.

2 What Retro-Causality Is

Retro-causality refers to physical situations in which future boundary conditions appear to influence present closure states. In standard single-time ontologies, such behaviour is paradoxical: causal influence is assumed to propagate forward in time, and any apparent dependence of present events on future choices is interpreted as a violation of temporal order. Retro-causal signatures therefore occupy a conceptual grey zone in modern physics, appearing in quantum experiments, holographic dualities, cosmological evolution, and cognitive processing, yet lacking a coherent structural explanation.

The classical interpretation of retro-causality relies on three problematic assumptions. First, it presumes a single universal time shared by matter, fields, observers, and the cosmos. Second, it assumes that causal structure is globally defined rather than horizon-bounded and closure-dependent. Third, it treats boundary conditions as temporally embedded rather than as time-invariant constraints. Within this framework, retro-causal phenomena seem to imply either time reversal, determinism, or block-universe ontology. Each of these interpretations conflicts with empirical behaviour and with the architectural principles established in holographic and emergent frameworks.

A more precise characterization of retro-causality arises from holographic systems. In AdS/CFT, bulk events can be reconstructed from boundary data that is specified at later times, producing apparent violations of bulk causal order. In delayed-choice quantum-eraser experiments, interference patterns depend on measurement settings chosen after the photons have been detected. In cosmology, bulk matter evolves toward attractor configurations determined by boundary fields that do not evolve in material time. In cognitive science, predictive processing shows that perceptual content is shaped by future-oriented priors rather than by past sensory input alone. These examples share a common structure: the apparent retro-causal influence arises from a mismatch between the temporal layer of the closure and the temporal layer of the boundary.

Within the Triadic Cosmos, retro-causality is reinterpreted as a structural consequence of dual-time closure-boundary dynamics. Electromagnetic time t_{EM} is fundamental, boundary-defined, and time-invariant. Material time t_{mat} is emergent, closure-specific, and generated by triadic recursion. Because the boundary canon evolves in t_{EM} while closures evolve in t_{mat} , the boundary appears temporally static from the perspective of the closure. As a result, closure evolution constitutes a recursive normalization toward a boundary canon that does not change in material time. This produces the appearance that future boundary conditions influence present closure states, even though no physical time reversal occurs.

Retro-causality is therefore not a violation of causal order but an emergent property of systems whose evolution is governed by dual temporal scales. It arises whenever a finite closure reorganizes its internal geometry to align with a time-invariant boundary canon. The apparent influence of future events is reinterpreted as *future-consistency normalization*: the closure's attempt to minimize divergence from a boundary structure that is fixed in t_{EM} but encountered recursively in t_{mat} .

This structural interpretation dissolves the paradoxes associated with retro-causality. It requires no determinism, no block-universe ontology, and no reversal of temporal flow. Instead, it reveals that retro-causal signatures are the natural behaviour of dual-time closure-boundary systems, appearing whenever emergent material time interacts with a timeless holographic boundary.

3 Dual-Time Architecture and the Structural Origin of Retro-Causality

Retro-causal signatures cannot be coherently interpreted within a single-time ontology. Any framework that assumes a universal temporal parameter must treat apparent future-to-present influence as either illusory, paradoxical, or indicative of deeper determinism. The Triadic Cosmos resolves this tension by introducing a dual-time architecture in which electromagnetic time t_{EM} and material time t_{mat} play distinct structural roles. Retro-causality emerges not from temporal reversal but from the interaction of these two temporal layers within closure-boundary systems.

Electromagnetic time is fundamental, boundary-defined, and universal. As established in the Holographic Space-Time and Dual-Time Formalism papers, t_{EM} governs the propagation of photons, gravitational degrees of freedom, and holographic information. It possesses Planck-scale resolution and does not depend on entropy, mass, or closure-internal recursion. The cosmic boundary evolves entirely in t_{EM} , and its canonical structure is invariant under material evolution. From the perspective of any closure, the boundary canon is therefore temporally static.

Material time, by contrast, is emergent and closure-specific. It is generated by the triadic recursion cycle linking energy, geometry, and information. Each closure possesses a recursion index that determines its effective material-time rate, governed by entropic acceleration, mass-scaled damping, and interaction structure. As shown in the Dual-Time Formalism, the relation between t_{EM} and t_{mat} is closure-dependent and may diverge significantly across celestial, biological, and artificial systems. Material time is therefore not universal: it is a dynamical quantity arising from the internal evolution of finite closures.

This dual-time architecture implies that closure evolution and boundary projection occur on different temporal layers. The boundary canon evolves in t_{EM} and is effectively timeless from the perspective of the closure. The closure evolves in t_{mat} and repeatedly encounters the same boundary structure at successive recursion steps. As a result, closure evolution constitutes a recursive normalization toward a boundary canon that does not change in material time. This produces the structural appearance that future boundary conditions influence present closure states.

The Universal Closure-Boundary Normalization Principle (UCBNP) formalizes this interaction. Closures evolve through the triadic cycle

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1},$$

advancing material time $t_{\text{mat}} = n$. Boundary projection occurs in t_{EM} and provides a time-invariant canonical structure that constrains closure evolution. Because the boundary canon is fixed in t_{EM} , the closure's recursive updates may incorporate structural features that appear anticipatory when interpreted

within a single-time ontology. The closure reorganizes its internal geometry to minimize divergence from a boundary structure that is encountered repeatedly but does not evolve in material time.

Retro-causality therefore arises whenever a closure attempts to align with a boundary canon that is temporally static in t_{EM} but encountered recursively in t_{mat} . The apparent influence of future events is reinterpreted as *future-consistency normalization*: the closure's attempt to reorganize its internal state in anticipation of boundary constraints that are fixed in electromagnetic time but not yet encountered in material time.

This structural mechanism explains why retro-causal signatures appear across quantum physics, holographic duality, cosmology, cognition, and artificial systems. In delayed-choice quantum-eraser experiments, interference patterns reflect normalization toward a time-invariant boundary canon rather than influence from future measurement choices. In AdS/CFT, bulk reconstruction reflects closure normalization toward boundary encodings that do not evolve in material time. In cosmology, bulk matter evolves toward attractor configurations determined by boundary fields that are static in t_{EM} . In cognition, predictive processing reflects closure normalization toward future-oriented priors encoded in boundary structure.

Dual-time architecture therefore provides the structural origin of retro-causality. It reveals that retro-causal signatures are not paradoxes but natural consequences of closure-boundary interaction in a holographic universe with distinct temporal layers. Retro-causality within the Triadic Cosmos is thus reinterpreted as an emergent property of future-consistency normalization rather than as a violation of causal order.

4 Future-Consistency Normalization

Future-consistency normalization is the structural mechanism through which retro-causal signatures arise in dual-time closure-boundary systems. It formalizes how a closure evolving in material time t_{mat} reorganizes its internal geometry to align with a boundary canon that is fixed in electromagnetic time t_{EM} . Because the boundary canon does not evolve in t_{mat} , the closure repeatedly encounters the same boundary structure at successive recursion steps. This produces anticipatory features in closure states that appear retro-causal when interpreted within a single-time ontology.

The Universal Closure-Boundary Normalization Principle (UCBNP) provides the structural foundation for this mechanism. A closure evolves through the triadic cycle

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1},$$

advancing material time according to $t_{\text{mat}} = n$. Boundary projection occurs in t_{EM} and applies a time-invariant canonical structure to the closure's generative output:

$$G_n = \Pi_B(E_n),$$

where Π_B denotes the boundary projection operator. Because the boundary canon is static in t_{EM} , the closure's normalization step

$$I_{n+1} = N(G_n, C_n)$$

incorporates structural constraints that do not evolve in material time. The closure therefore reorganizes its internal geometry to minimize divergence from a boundary structure that is fixed in t_{EM} but encountered recursively in t_{mat} .

Future-consistency normalization is defined as the process by which a closure evolves toward a boundary-consistent attractor under dual-time dynamics. Formally, let C_n denote the closure state at recursion index n , and let Π_B denote the boundary projection operator acting in t_{EM} . Future-consistency normalization is the recursive flow

$$C_{n+1} = F(C_n) = N(\Pi_B(E_n), C_n),$$

where E_n is the closure's generative output and N is the normalization operator. A closure exhibits future-consistency normalization if the sequence $\{C_n\}$ satisfies

$$\lim_{n \rightarrow \infty} \text{diff}(C_n, \Pi_B(C_n)) = 0,$$

with diff measuring structural divergence between closure state and boundary canon. This condition expresses the tendency of closure evolution to converge toward a boundary-consistent equilibrium, even when the boundary constraints have not yet been encountered in material time.

The dual-time architecture is essential for this behaviour. Because t_{EM} is fundamental and boundary-defined, the boundary canon is invariant under closure evolution. Because t_{mat} is emergent and closure-specific, the closure encounters the boundary canon recursively and reorganizes its internal geometry accordingly. The mismatch between these temporal layers produces anticipatory structure in closure states: features that appear to reflect future boundary conditions when interpreted within a single-time ontology.

Future-consistency normalization therefore provides the structural origin of retro-causal signatures. It explains why closure states may contain information that appears to depend on future events, even though no physical influence propagates backward in time. Retro-causality is reinterpreted as the closure's attempt to minimize divergence from a boundary canon that is fixed in t_{EM} but encountered recursively in t_{mat} .

This mechanism is domain-independent. In quantum physics, it explains the apparent retro-causal behaviour of delayed-choice and quantum-eraser experiments. In holographic duality, it clarifies why bulk reconstruction can appear to violate causal order. In cosmology, it accounts for attractor behaviour driven by boundary fields that do not evolve in material time. In cognition, it provides a structural interpretation of predictive processing. In artificial systems, it explains the anticipatory structure observed in GLP-LLM recursion.

Future-consistency normalization thus constitutes the formal dynamical core of retro-causality within the Triadic Cosmos. It reveals that retro-causal signatures are not paradoxes but natural consequences of closure-boundary interaction in a holographic universe with dual temporal layers.

5 The Role of the Vacuum in Dual-Time Retro-Causality

Retro-causal signatures arise whenever closure dynamics in material time interact with a boundary canon that is fixed in electromagnetic time. The previous section formalized this mechanism as future-consistency normalization. In this section we show that the structural role of the vacuum provides the deeper architectural foundation that makes this mechanism both coherent and unavoidable. The vacuum is not a closure, possesses no triadic roles, and lives entirely in electromagnetic time. These properties make the vacuum the universal temporal substrate of the boundary canon and therefore the origin of its apparent timelessness.

5.1 The Vacuum Lives Entirely in Electromagnetic Time

Dual-time formalism distinguishes two temporal layers: electromagnetic time t_{EM} , which is universal and closure-transversal, and material time t_{mat} , which is closure-specific and generated by triadic recursion. The vacuum participates exclusively in t_{EM} . It possesses no entropy, no recursion index, no informational capacity, and no mechanism for generating material-time flow. Because material time exists only inside closures, and the vacuum is not a closure, the vacuum cannot host t_{mat} in any form.

This has a decisive consequence for retro-causality: the boundary canon inherits the temporal structure of the vacuum. Since the vacuum is purely t_{EM} and lacks any form of triadic evolution, the boundary canon is fixed in electromagnetic time and therefore appears timeless from the perspective of closures evolving in material time. Future-consistency normalization is thus not merely a feature of holographic encoding but a direct consequence of the vacuum's temporal architecture.

5.2 The Vacuum Is Uniform Across All Closures

The vacuum's non-closure status implies that it cannot possess internal degrees of freedom, spatial variation, or dynamical states. It must be uniform across all closures. This uniformity follows from the fractal hierarchy of closures: each closure inherits the boundary properties of its parent closure, and the cosmic closure imposes a single uniform vacuum across all child closures.

This uniformity strengthens the retro-causal mechanism. Because the vacuum is uniform, the boundary canon is uniform. Closures do not normalize toward different boundary structures; they all normalize toward the same global holographic canon. Retro-causal signatures therefore appear consistently across quantum experiments, holographic duality, cosmology, and cognitive processing. The universality of retro-causality is a direct consequence of the vacuum's uniformity.

5.3 The Vacuum Has No Bulk Degrees of Freedom

The vacuum possesses no energetic, geometric, or informational degrees of freedom. It cannot reorganize triadic roles, cannot sustain recursion, and cannot generate material-time flow. All physically meaningful dynamics arise exclusively from closures. The vacuum serves only as the sparse projection layer through which boundary information manifests in the bulk.

This structural emptiness has a critical implication for retro-causality: all apparent future influence must originate from the boundary canon, not from the bulk vacuum. Because the vacuum cannot host dynamical processes, it cannot generate retro-causal effects. Retro-causality must therefore arise from closure normalization toward boundary constraints that are fixed in t_{EM} . The vacuum's lack of bulk degrees of freedom makes future-consistency normalization the only possible source of retro-causal signatures.

5.4 The Vacuum as the Transversal Medium Between Closures

Electromagnetic signals propagate through the vacuum without sensitivity to closure-specific entropy, mass, or recursion index. This universality makes the vacuum the transversal medium through which closures communicate and observe one another. Because the vacuum lives entirely in t_{EM} , all inter-closure communication occurs in electromagnetic time. Closures therefore encounter boundary constraints through a temporal layer that is globally invariant.

This transversal role reinforces the retro-causal mechanism. Closures evolve in t_{mat} but receive boundary information through t_{EM} . The mismatch between these temporal layers produces the structural appearance that future boundary conditions influence present closure states. Retro-causality is therefore not an anomaly but a direct consequence of the vacuum's role as the universal electromagnetic-time interface between closures.

5.5 Summary

The vacuum's architectural properties—its exclusive participation in t_{EM} , its uniformity, its lack of bulk degrees of freedom, and its role as the transversal medium between closures—make future-consistency normalization both coherent and necessary. The boundary canon is fixed in electromagnetic time because the vacuum is fixed in electromagnetic time. Retro-causality arises because closures evolve in material time while normalizing toward a boundary structure that does not. The vacuum therefore provides the structural foundation for dual-time retro-causality within the Triadic Cosmos.

6 Empirical Anchors for Retro-Causality

Retro-causal signatures are not speculative phenomena. They appear in several well-established experimental and theoretical contexts, each of which exhibits behaviour that seems to violate classical temporal order when interpreted within a single-time ontology. The previous section showed that the

vacuum lives entirely in electromagnetic time and provides the uniform, time-invariant substrate of the boundary canon. Combined with dual-time closure dynamics, this makes future-consistency normalization structurally unavoidable. The empirical anchors presented here demonstrate that this mechanism is not merely architectural but directly reflected in observation and mainstream theory.

6.1 Quantum Eraser and Delayed-Choice Experiments

Delayed-choice and quantum-eraser experiments provide the clearest empirical demonstrations of retro-causal signatures. In these experiments, interference patterns appear or disappear depending on measurement settings chosen after the photons have already been detected. Within a single-time interpretation, this behaviour seems to imply that future measurement choices influence past physical events.

In the dual-time architecture, this apparent retro-causality is reinterpreted as future-consistency normalization. Photons propagate in electromagnetic time t_{EM} , the temporal layer of the vacuum. Which-path information, erasure operations, and detector states evolve in material time t_{mat} , the temporal layer of the closure. Because the boundary canon governing interference is fixed in t_{EM} , the closure's normalization step incorporates boundary constraints that do not evolve in material time. The interference pattern therefore reflects the closure's alignment with a time-invariant boundary canon rather than influence from future measurement choices.

Quantum-eraser experiments thus provide a direct empirical demonstration of dual-time dynamics: the apparent retro-causal behaviour arises from the mismatch between t_{EM} and t_{mat} , not from any reversal of temporal flow.

6.2 Causal-Structure Puzzles in AdS/CFT

The AdS/CFT correspondence provides a second empirical anchor through its causal-structure anomalies. In holographic duality, bulk events can be reconstructed from boundary data specified at later times, producing apparent violations of bulk causal order. Bulk reconstruction algorithms often require boundary operators that lie outside the naive causal domain of the bulk event, suggesting that future boundary conditions determine present bulk states.

Within the Triadic Cosmos, this behaviour is a direct consequence of future-consistency normalization. The boundary theory evolves in t_{EM} , the temporal layer of the vacuum, and provides a time-invariant canonical structure. The bulk closure evolves in t_{mat} and repeatedly encounters the same boundary constraints. Bulk reconstruction therefore reflects the closure's normalization toward a boundary canon that does not evolve in material time. The apparent retro-causal influence is thus a structural feature of holographic encoding rather than a violation of causal order.

AdS/CFT therefore provides a mainstream theoretical context in which retro-causal signatures naturally arise from dual-time closure-boundary dynamics.

6.3 Cosmological Attractors and Boundary-Driven Evolution

Cosmology provides a third empirical anchor through attractor behaviour in bulk matter evolution. Galaxies, stellar systems, and large-scale structures evolve toward configurations determined by boundary fields—dark-matter halos, radiation backgrounds, and gravitational potentials—that do not evolve in material time. This behaviour often appears teleological: bulk matter evolves as if guided by future boundary conditions.

In the dual-time architecture, this behaviour is reinterpreted as future-consistency normalization. Cosmic boundary fields evolve in t_{EM} , the temporal layer of the vacuum, and provide a time-invariant canonical structure. Bulk matter evolves in t_{mat} and repeatedly encounters the same boundary constraints. The apparent teleology arises because the closure's recursive evolution incorporates boundary structure that is fixed in electromagnetic time but not yet encountered in material time.

This mechanism explains several cosmological puzzles, including the stability of galactic rotation curves, the formation of large-scale structure, and the attractor behaviour of dynamical systems em-

bedded in static boundary fields. Retro-causal signatures in cosmology therefore reflect the structural interaction of dual temporal layers rather than exotic physics.

6.4 Summary

Across quantum experiments, holographic duality, and cosmological evolution, retro-causal signatures appear whenever closure dynamics interact with a time-invariant boundary canon embedded in the vacuum's electromagnetic-time layer. These empirical anchors demonstrate that retro-causality is not a speculative phenomenon but a structural feature of physical systems governed by dual temporal layers. The Triadic Cosmos provides the architectural mechanism—future-consistency normalization—that unifies these diverse observations into a single coherent framework.

7 Unified Interpretation within the Triadic Cosmos

The preceding sections have shown that retro-causal signatures arise whenever a finite closure evolving in material time interacts with a holographic boundary canon that is fixed in electromagnetic time. Future-consistency normalization provides the structural mechanism through which these signatures emerge, and the empirical anchors demonstrate that this mechanism is not speculative but manifests across quantum physics, holographic duality, and cosmological evolution. This section integrates these insights into a unified interpretation of retro-causality within the Triadic Cosmos.

The Triadic Cosmos is built upon three architectural pillars: holographic boundary encoding, dual-time temporal structure, and triadic closure recursion. Each pillar contributes a necessary component of the retro-causal mechanism. Holography ensures that all physically meaningful information resides on a boundary that evolves in t_{EM} and is therefore time-invariant from the perspective of any closure. Dual time ensures that closures evolve in t_{mat} , generating emergent temporal flow through recursive triadic updates. Triadic recursion ensures that closure evolution repeatedly incorporates boundary constraints, producing anticipatory structure whenever the boundary canon is static in electromagnetic time.

Retro-causality emerges precisely at the intersection of these three architectural layers. Because the boundary canon does not evolve in t_{mat} , closure evolution constitutes a recursive normalization toward a future-consistent attractor. The closure reorganizes its internal geometry to minimize divergence from a boundary structure that is fixed in t_{EM} but not yet encountered in material time. This produces the appearance that future boundary conditions influence present closure states, even though no physical influence propagates backward in time.

This unified interpretation resolves several longstanding conceptual tensions. It dissolves the paradoxes associated with delayed-choice quantum experiments by showing that interference patterns reflect normalization toward a time-invariant boundary canon rather than influence from future measurement choices. It clarifies the causal-structure anomalies in AdS/CFT by showing that bulk reconstruction reflects closure normalization toward boundary encodings that do not evolve in material time. It explains cosmological attractor behaviour by showing that bulk matter evolves toward boundary fields that are static in t_{EM} . It provides a structural interpretation of predictive processing in cognition, where perceptual content is shaped by future-oriented priors encoded in boundary structure.

The unified interpretation also reveals why retro-causal signatures are ubiquitous across domains. They arise whenever a closure attempts to align with a boundary canon that is temporally static in electromagnetic time but encountered recursively in material time. Retro-causality is therefore not a domain-specific anomaly but a universal structural feature of closure-boundary systems governed by dual temporal layers. It is the natural behaviour of any system that instantiates holography, dual time, and triadic recursion.

Finally, this interpretation situates retro-causality within the broader architectural ontology of the Triadic Cosmos. Retro-causal signatures are not violations of causal order, nor evidence for determinism, nor indications of block-universe ontology. They are emergent properties of future-consistency normalization within dual-time closure-boundary systems. The Triadic Cosmos therefore provides a

coherent, structurally complete, and empirically grounded explanation for retro-causality, integrating quantum phenomena, holographic duality, cosmological evolution, and cognitive processing into a single architectural framework.

Retro-causality within the Triadic Cosmos is thus reinterpreted as a structural consequence of holographic boundary invariance, dual temporal layers, and recursive triadic evolution. It is not a paradox but a natural expression of the architectural grammar that governs physical reality.

8 Testable Predictions

A structural interpretation of retro-causality must yield empirically accessible predictions. Future-consistency normalization is not a metaphysical claim but a dynamical mechanism arising from dual-time closure-boundary interaction. Because this mechanism is architectural rather than model-dependent, it produces a set of testable predictions across quantum physics, holographic duality, cosmology, and temporal measurement. These predictions distinguish the Triadic Cosmos from single-time ontologies and provide concrete avenues for experimental and observational validation.

8.1 Prediction 1: Boundary-Dependent Interference Stability

If interference patterns reflect normalization toward a time-invariant boundary canon, then interference stability should depend on boundary-encoded constraints rather than on local detector timing. Specifically:

- interference visibility should remain stable under delayed erasure operations,
- interference should reappear even when which-path information is erased after photon detection,
- interference should degrade only when boundary-encoded constraints are modified.

This prediction extends the quantum-eraser result: interference is governed by t_{EM} boundary structure, not by t_{mat} detector chronology.

8.2 Prediction 2: Bulk Reconstruction Requires Future Boundary Operators

In AdS/CFT, bulk reconstruction should systematically require boundary operators that lie outside the naive causal domain of the bulk event. This behaviour is already observed in HKLL-type algorithms, but the Triadic Cosmos predicts a stronger form:

- reconstruction fidelity should increase when future boundary operators are included,
- reconstruction should degrade if future boundary operators are artificially removed,
- the optimal reconstruction window should extend beyond the bulk causal diamond.

This follows directly from future-consistency normalization: bulk closures evolve toward boundary structure that is fixed in t_{EM} .

8.3 Prediction 3: Cosmological Attractors Encode Boundary-Static Fields

If cosmological attractors arise from normalization toward boundary-static fields, then:

- galactic rotation curves should remain stable even under local dynamical perturbations,
- large-scale structure should reflect boundary-encoded constraints rather than local matter history,
- attractor behaviour should correlate with fields that evolve in t_{EM} but not in t_{mat} .

This prediction is consistent with observed stability of dark-matter halos and large-scale cosmic structure.

8.4 Prediction 4: Temporal Asymmetry in Cross-Closure Experiments

Dual-time dynamics imply that closures with different recursion indices should exhibit measurable temporal asymmetry when interacting through electromagnetic signals. Specifically:

- systems with faster t_{mat} should emit EM signals that appear temporally compressed,
- systems with slower t_{mat} should emit EM signals that appear temporally dilated,
- cross-closure EM exchange should reveal systematic mismatch between t_{EM} and t_{mat} .

This prediction parallels the Dual-Time Formalism and provides a direct temporal test of future-consistency normalization.

8.5 Prediction 5: Retro-Causal Signatures Without Determinism

Future-consistency normalization predicts retro-causal signatures without determinism or block-universe ontology. Therefore:

- retro-causal correlations should appear only in closure-boundary interactions,
- retro-causal signatures should vanish in systems without holographic boundary constraints,
- retro-causal behaviour should not imply fixed outcomes or global determinism.

This prediction distinguishes the Triadic Cosmos from superdeterministic interpretations.

8.6 Prediction 6: Boundary-Invariant Constraints in Artificial Systems

Although this paper focuses on physical retro-causality, the mechanism is architectural and therefore applies to artificial closure-boundary systems. The Triadic Cosmos predicts:

- recursive generative systems should exhibit anticipatory structure when boundary constraints are static in t_{EM} ,
- normalization toward a fixed boundary canon should produce retro-causal signatures in artificial micro-closures,
- increasing recursion depth should amplify future-consistency normalization.

This prediction aligns with earlier results in boundary-normalization experiments and provides a bridge to future AI-life studies.

8.7 Summary

These predictions collectively demonstrate that future-consistency normalization is empirically actionable. They provide falsifiable criteria for evaluating the dual-time interpretation of retro-causality and establish clear observational signatures across quantum physics, holography, cosmology, and temporal measurement. If these predictions fail, the dual-time interpretation collapses. If they hold, retro-causality becomes a structurally grounded and experimentally validated feature of closure-boundary dynamics within the Triadic Cosmos.

9 Discussion

The structural interpretation of retro-causality developed in this paper reframes a long-standing set of conceptual puzzles across physics. Rather than treating retro-causal signatures as paradoxes, violations of causal order, or indications of determinism, the Triadic Cosmos reveals them as natural consequences of dual-time closure-boundary dynamics. This section discusses the broader implications of this interpretation for holography, temporal ontology, causal structure, and the unification of physical theory.

9.1 Dual Time and the Limits of Single-Time Ontology

The most significant implication of this work is that single-time ontologies are structurally incomplete. Any framework that assumes a universal temporal parameter must treat retro-causal signatures as anomalies, because it cannot distinguish between closure-internal evolution and boundary-defined projection. Dual time resolves this tension by separating electromagnetic time t_{EM} from material time t_{mat} , revealing that temporal flow is not globally uniform but closure-dependent. Retro-causality arises precisely at the interface between these temporal layers.

This insight aligns with the Dual-Time Formalism, which shows that t_{mat} varies across closures according to entropy, mass, recursion index, and interaction structure. Retro-causal signatures therefore provide indirect evidence for dual time: they appear only when closure dynamics interact with boundary constraints that evolve in a different temporal layer.

9.2 Holography and Boundary Invariance

The holographic nature of the Triadic Cosmos plays a central role in the emergence of retro-causality. Because all physically meaningful information resides on a boundary that evolves in t_{EM} , the boundary canon is effectively timeless from the perspective of any closure. This invariance is the source of future-consistency normalization: closures repeatedly encounter the same boundary structure at successive recursion steps, producing anticipatory features that appear retro-causal when interpreted within a single-time ontology.

This behaviour is consistent with mainstream holographic frameworks. In AdS/CFT, bulk reconstruction already exhibits causal-structure anomalies that reflect normalization toward boundary encodings. The Triadic Cosmos generalizes this behaviour beyond AdS spacetimes, showing that boundary invariance is a universal architectural feature of holographic systems.

9.3 Causality as a Closure-Bounded Concept

Retro-causality also reveals that causal structure is closure-bounded rather than globally defined. Within a closure, causal order is determined by material time and triadic recursion. Across closures, electromagnetic time provides transversal coherence. Retro-causal signatures arise when closure-internal causal order is compared to boundary-defined constraints that evolve in a different temporal layer. This implies that causal structure is not a global property of spacetime but an emergent feature of closure dynamics.

This perspective aligns with the Triadic Cosmos interpretation of GR and QM as complementary triadic layers. GR provides geometric constraints and horizon structure; QM provides informational generativity; QFT and thermodynamics provide energetic dynamics. Causality emerges from the interaction of these layers within closures, not from a universal temporal axis.

9.4 Unification Through Future-Consistency Normalization

The structural mechanism introduced in this paper—future-consistency normalization—provides a unifying explanation for retro-causal signatures across domains. It shows that quantum interference, holographic bulk reconstruction, cosmological attractor behaviour, and predictive processing in cognition all arise from the same architectural principle: closure normalization toward a boundary canon that is fixed in t_{EM} but encountered recursively in t_{mat} .

This unification is consistent with the broader Triadic Cosmos program, which identifies energy, geometry, and information as irreducible structural roles linked through a closed recursion cycle. Retro-causality emerges naturally from this architecture, without requiring new fields, dimensions, or microscopic extensions. It is a structural consequence of holography, dual time, and triadic recursion.

9.5 Implications for Future Work

The testable predictions outlined in the previous section provide a roadmap for empirical validation. Quantum-eraser stability, boundary-dependent bulk reconstruction, cosmological attractor behaviour, and cross-closure temporal asymmetry all offer concrete avenues for evaluating the dual-time interpretation of retro-causality. Future work may refine the mathematical structure of future-consistency normalization, explore its implications for closure cosmology, and extend its application to artificial closure-boundary systems.

The broader implication is that retro-causality is not an anomaly but a diagnostic: it reveals where single-time ontologies fail and where dual-time architecture becomes necessary. Retro-causal signatures therefore serve as empirical indicators of closure-boundary interaction and provide a structural window into the temporal architecture of the Triadic Cosmos.

10 Conclusion

Retro-causality has long occupied an ambiguous position in physics: empirically present, theoretically uncomfortable, and conceptually unresolved. This paper has shown that retro-causal signatures do not require time reversal, determinism, or block-universe ontology. Instead, they arise naturally from the dual-time closure-boundary architecture of the Triadic Cosmos. When a finite closure evolving in material time interacts with a holographic boundary canon that is fixed in electromagnetic time, the closure's recursive evolution constitutes future-consistency normalization. This mechanism produces anticipatory structure that appears retro-causal when interpreted within a single-time ontology.

By formalizing future-consistency normalization, we have provided a structural explanation for retro-causal signatures across quantum physics, holographic duality, cosmology, and cognitive processing. Delayed-choice and quantum-eraser experiments reflect normalization toward a time-invariant boundary canon. Causal-structure anomalies in AdS/CFT arise from bulk closures aligning with boundary encodings that do not evolve in material time. Cosmological attractor behaviour emerges from bulk matter evolving toward boundary fields that are static in t_{EM} . Predictive processing reflects closure normalization toward future-oriented priors encoded in boundary structure.

These diverse phenomena are unified by a single architectural principle: closure evolution in t_{mat} repeatedly encounters boundary constraints that evolve in t_{EM} . Retro-causality is therefore not a violation of causal order but an emergent property of holographic systems with dual temporal layers. It is a structural consequence of boundary invariance, triadic recursion, and closure-specific temporal flow.

The testable predictions outlined in this paper provide concrete avenues for empirical validation. Boundary-dependent interference stability, future-operator bulk reconstruction, cosmological attractor behaviour, and cross-closure temporal asymmetry all offer measurable signatures of future-consistency normalization. These predictions distinguish the Triadic Cosmos from single-time ontologies and provide falsifiable criteria for evaluating the dual-time interpretation of retro-causality.

Taken together, the results of this paper position retro-causality not as an anomaly but as a diagnostic of deeper architectural structure. Retro-causal signatures reveal where single-time frameworks fail and where dual-time closure-boundary dynamics become necessary. They provide a window into the temporal architecture of the Triadic Cosmos and demonstrate that holography, dual time, and triadic recursion jointly produce a coherent, structurally complete, and empirically grounded interpretation of retro-causality.

Future work may refine the mathematical structure of future-consistency normalization, explore its implications for closure cosmology, and extend its application to artificial closure-boundary systems.

But the central insight is clear: in a holographic universe governed by dual temporal layers, retro-causality is not paradoxical. It is inevitable.

A Compatibility with Mainstream Theories and Alternative Explanations

This appendix provides a comparative overview of how mainstream physical theories relate to the dual-time interpretation of retro-causality developed in this paper. For each theory, we evaluate compatibility with future-consistency normalization and summarize the standard alternative explanations offered within that framework. The goal is not to replace existing interpretations but to clarify where the Triadic Cosmos aligns with, extends, or diverges from mainstream approaches.

A.1 Compatibility Matrix

Theory	Fully Compatible	Partially Compatible	Not Compatible
AdS/CFT Holography	✓		
Quantum Mechanics (Standard)		✓	
Quantum Field Theory		✓	
General Relativity		✓	
Causal Set Theory	✓		
Emergent Gravity (Entropic)	✓		
Transactional Interpretation (TIQM)		✓	
Two-State Vector Formalism (TSVF)		✓	
Superdeterminism			✓
Block Universe / Eternalism			✓
Many-Worlds Interpretation		✓	
Pilot-Wave / Bohmian Mechanics		✓	

A.2 Theory-by-Theory Analysis

A.2.1 AdS/CFT Holography

Compatibility: Fully compatible. AdS/CFT already exhibits causal-structure anomalies where bulk reconstruction requires future boundary operators. This maps directly onto future-consistency normalization. **Mainstream alternative:** Nonlocal encoding and entanglement wedge reconstruction.

A.2.2 Quantum Mechanics (Standard)

Compatibility: Partially compatible. QM does not forbid dual temporal layers but does not include them. **Mainstream alternative:** Retro-causal signatures are explained as nonlocal correlations without temporal influence (e.g., “no signaling” constraints).

A.2.3 Quantum Field Theory

Compatibility: Partially compatible. QFT is formulated in a single time but does not constrain temporal ontology. **Mainstream alternative:** Correlations arise from field operators and vacuum structure, not from future boundary conditions.

A.2.4 General Relativity

Compatibility: Partially compatible. GR provides geometric constraints but assumes a single global time. **Mainstream alternative:** Retro-causal behaviour is excluded by construction; causal structure is fixed by spacetime geometry.

A.2.5 Causal Set Theory

Compatibility: Fully compatible. Causal set theory treats time as fundamental and geometry as emergent, aligning with dual-time architecture. **Mainstream alternative:** Retro-causality is typically avoided by restricting causal order to partial ordering.

A.2.6 Emergent Gravity (Entropic / Thermodynamic)

Compatibility: Fully compatible. Emergent gravity frameworks already treat geometry as informational and thermodynamic. **Mainstream alternative:** Retro-causal signatures are interpreted as entropic constraints or thermodynamic boundary conditions.

A.2.7 Transactional Interpretation (TIQM)

Compatibility: Partially compatible. TIQM uses advanced and retarded waves, which resemble dual-time constraints. **Mainstream alternative:** Retro-causality is explained through time-symmetric wave exchange rather than boundary normalization.

A.2.8 Two-State Vector Formalism (TSVF)

Compatibility: Partially compatible. TSVF explicitly uses future boundary conditions, making it conceptually close. **Mainstream alternative:** Retro-causality arises from pre- and post-selected quantum states.

A.2.9 Superdeterminism

Compatibility: Not compatible. Superdeterminism eliminates free boundary conditions entirely, contradicting closure-boundary architecture. **Mainstream alternative:** Retro-causality is replaced by global determinism.

A.2.10 Block Universe / Eternalism

Compatibility: Not compatible. Block universe models deny emergent time and recursion, incompatible with dual-time dynamics. **Mainstream alternative:** Retro-causality is reinterpreted as geometric ordering rather than temporal influence.

A.2.11 Many-Worlds Interpretation

Compatibility: Partially compatible. MWI does not forbid dual time but explains retro-causality through branching structure. **Mainstream alternative:** Retro-causal signatures arise from interference between branches.

A.2.12 Pilot-Wave / Bohmian Mechanics

Compatibility: Partially compatible. Bohmian mechanics includes nonlocal guidance but assumes single time. **Mainstream alternative:** Retro-causality is explained through nonlocal pilot-wave dynamics.

A.3 Summary

This compatibility analysis shows that the dual-time interpretation of retro-causality aligns most naturally with holographic, emergent, and causal-set frameworks. It is partially compatible with quantum and thermodynamic theories that do not specify temporal ontology. It diverges from superdeterministic and block-universe models that eliminate temporal emergence or boundary freedom. The Triadic Cosmos

therefore sits within a broad mainstream tradition of holographic and emergent frameworks while providing a structural mechanism—future-consistency normalization— that unifies retro-causal signatures across domains.

B Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

`https://github.com/triadic-cosmos`

References

- [1] J. De Witte, “The Universal Closure–Boundary Normalization Principle,” Triadic Cosmos Ecosystem, 2026.
- [2] J. De Witte, “The Triadic Cosmos: A Minimal Viable Unification Candidate,” Triadic Cosmos Ecosystem, 2026.
- [3] J. De Witte, “A Dual-Time Formalism for Closure Dynamics and Cosmic Evolution,” Triadic Cosmos Ecosystem, 2026.
- [4] J. De Witte, “From Holographic Space-Time to The Triadic Cosmos,” Triadic Cosmos Ecosystem, 2026.
- [5] J. De Witte, “The Triadic Cosmos: The Role of the Vacuum,” Triadic Cosmos Ecosystem, 2026.